

Task 1: Record All Commands

Git, GitHub and Jenkins CI/CD Experiment Evidence

Student	Amit Mohanty
Roll No.	2301730325
GitHub Repository	devops-git-lab
Main Branch	master
Feature Branch	feature-update
Jenkins Pipeline	devops-git-lab-pipeline

Objective: Record the commands, repository operations, commit history, Jenkins configuration and pipeline execution evidence for the DevOps experiment.

1. Experiment Summary

The experiment demonstrates a basic Git-based workflow followed by Jenkins CI/CD. The repository was initialized locally, a README file was committed, a feature branch was created and updated, the change was merged into master, and both branches were pushed to GitHub. Jenkins was then configured to retrieve the repository and execute the Jenkinsfile.

Git setup	Version check, user configuration and repository initialization
Version control	Add, commit, branch, checkout, merge and log
Remote repository	git remote and git push to GitHub
Jenkins	Installation, job configuration and SCM settings
Pipeline	Jenkinsfile with Build, Test and Deploy stages
Execution	Successful Jenkins builds and console output

2. Git Commands and Version Control

Evidence 1: Git version verification using git --version.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1
$ git --version
git version 2.53.0.windows.2
```

Evidence 2: Git configuration: username and email configured using git config.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1
$ git config --global user.name "AmitMohanty"

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1
$ git config --global user.email "amitmohantyxiv@gmail.com"
```

Evidence 3: Project folder creation and repository initialization using git init.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1
• $ mkdir devops-git-lab

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1
• $ cd devops-git-lab

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab
• $ git init
Initialized empty Git repository in C:/Users/Amit/Desktop/devops stuff/lab 1/devops-git-lab/.git/
```

Evidence 4: README file creation and git status showing the untracked README.txt.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ echo "First Devops Lab" > README.txt

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ ls
README.txt

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Evidence 5: Staging and initial commit, followed by git log --oneline.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git add README.txt
warning: in the working copy of 'README.txt', LF will be replaced by CRLF the next time Git touches it

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.txt

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git commit -m "Add intial README file"
[master (root-commit) aef515a] Add intial README file
 1 file changed, 1 insertion(+)
 create mode 100644 README.txt

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git log --oneline
aef515a (HEAD -> master) Add intial README file
```

Evidence 6: Feature branch creation and checkout using git branch and git checkout.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git branch feature-update

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git branch
  feature-update
* master

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git checkout feature-update
Switched to branch 'feature-update'

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
○ $
```

Evidence 7: Feature branch file modification, staging and commit.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ echo "This line was added from the feature branch.">>README.txt

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ cat README.txt
First Devops Lab
This line was added from the feature branch.

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.txt

no changes added to commit (use "git add" and/or "git commit -a")

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ git add README.txt
warning: in the working copy of 'README.txt', LF will be replaced by CRLF the next time Git touches it

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ git commit -m "Update README from feature branch"
[feature-update 3ab822c] Update README from feature branch
 1 file changed, 1 insertion(+)

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ git log --oneline
3ab822c (HEAD -> feature-update) Update README from feature branch
aef515a (master) Add intial README file
```

Evidence 8: Checkout to master and fast-forward merge of feature-update, verified with git log --graph.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (feature-update)
• $ git checkout master
Switched to branch 'master'

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git merge feature-update
Updating aef515a..3ab822c
Fast-forward
 README.txt | 1 +
 1 file changed, 1 insertion(+)

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ cat README.txt
First Devops Lab
This line was added from the feature branch.

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git log --oneline --all --graph --decorate
* 3ab822c (HEAD -> master, feature-update) Update README from feature branch
* aef515a Add intial README file
```

Evidence 9: Git remote configuration and push of master and feature-update branches to GitHub.

```
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git remote add origin https://github.com/Amit-M-14/devops-git-lab..git

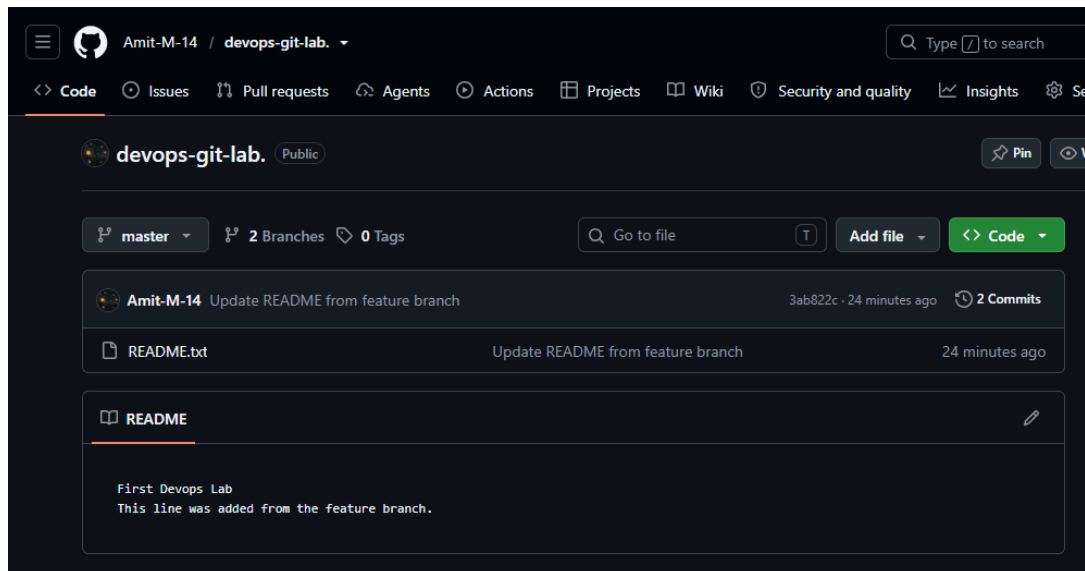
Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git remote -v
origin https://github.com/Amit-M-14/devops-git-lab..git (fetch)
origin https://github.com/Amit-M-14/devops-git-lab..git (push)

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 542 bytes | 36.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Amit-M-14/devops-git-lab..git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Amit@DESKTOP-6DCNV66 MINGW64 ~/Desktop/devops stuff/lab 1/devops-git-lab (master)
• $ git push -u origin feature-update
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:   https://github.com/Amit-M-14/devops-git-lab./pull/new/feature-update
remote:
To https://github.com/Amit-M-14/devops-git-lab..git
 * [new branch]      feature-update -> feature-update
branch 'feature-update' set up to track 'origin/feature-update'.
```

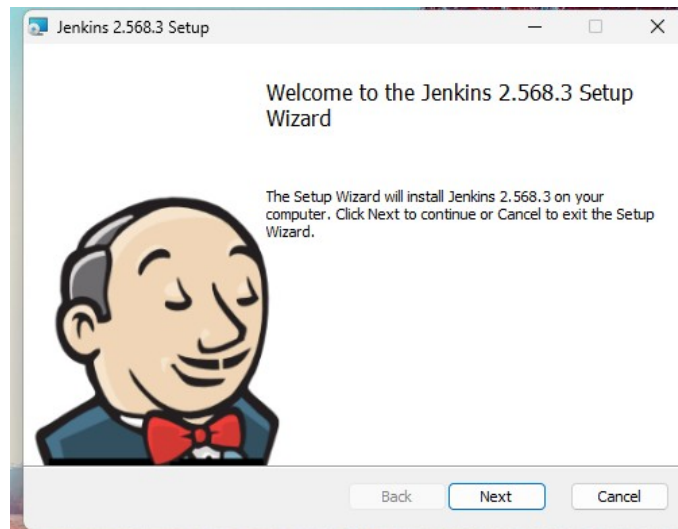
3. GitHub Repository

Evidence 10: GitHub repository showing README.txt and repository history.

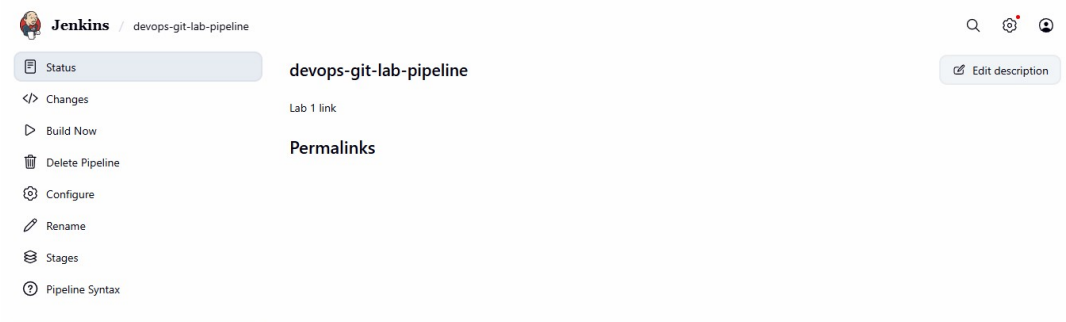


4. Jenkins CI/CD Evidence

Evidence 11: Jenkins installation setup wizard.



Evidence 12: Jenkins pipeline job overview after creation.



Evidence 13: Jenkins pipeline job general configuration.

Jenkins

devops-git-lab-pipeline

Configure

Search

Settings

User

Configure

General

Triggers

Pipeline

Advanced

General

Description

Enabled

Lab 1 link

Plain text

Preview

☐

Discard old builds

☐

Do not allow concurrent builds

☐

Do not allow the pipeline to resume if the controller restarts

☐

GitHub project

☐

Pipeline speed/durability override

☐

Preserve stashes from completed builds

Save

Apply

Evidence 14: Jenkins SCM configuration — Repository URL and Credentials.



This screenshot shows the Jenkins SCM configuration interface. The 'Repository URL' field contains the text 'https://github.com/Amit-M-14/devops-git-lab.'. Below it, the 'Credentials' dropdown menu is set to '- none -', with an '+ Add' button to its right. At the bottom of the configuration area, there is an 'Advanced' button with a downward arrow.

Repository URL ?

https://github.com/Amit-M-14/devops-git-lab.

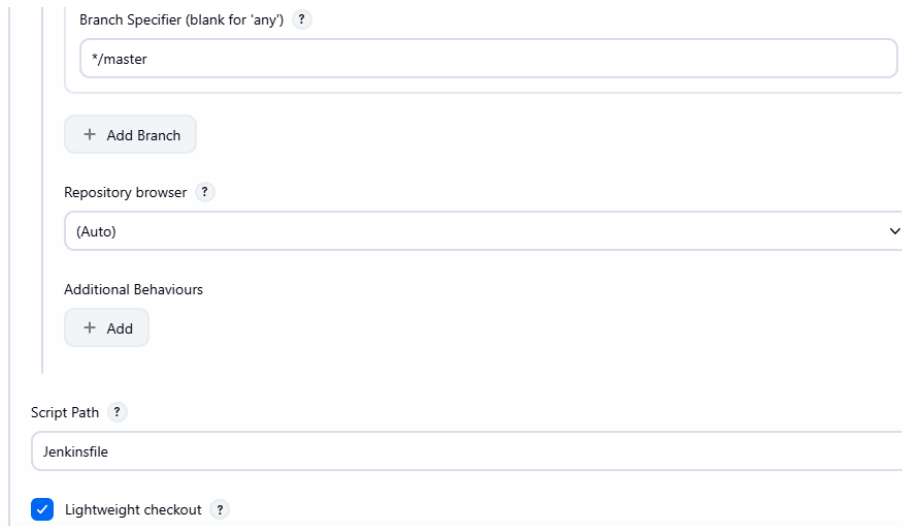
Credentials ?

- none -

+ Add

Advanced ▾

Evidence 15: Jenkins SCM configuration — Branch Specifier and Script Path.



This screenshot shows the Jenkins SCM configuration interface. The 'Branch Specifier (blank for 'any')' field contains the text '*/master'. Below this field is a '+ Add Branch' button. The 'Repository browser' dropdown menu is set to '(Auto)'. Under the 'Additional Behaviours' section, there is a '+ Add' button. The 'Script Path' field contains the text 'Jenkinsfile'. At the bottom, the 'Lightweight checkout' checkbox is checked.

Branch Specifier (blank for 'any') ?

*/master

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

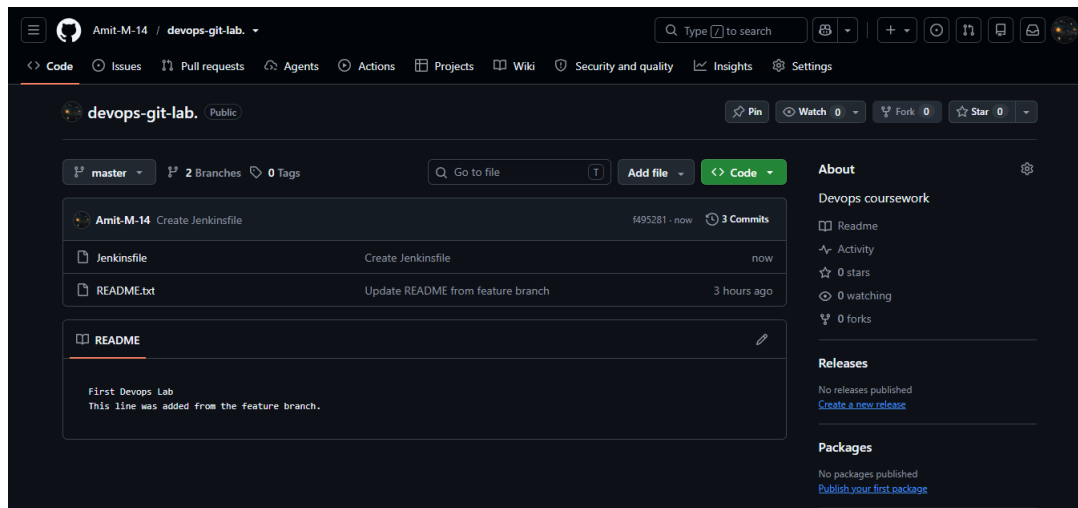
+ Add

Script Path ?

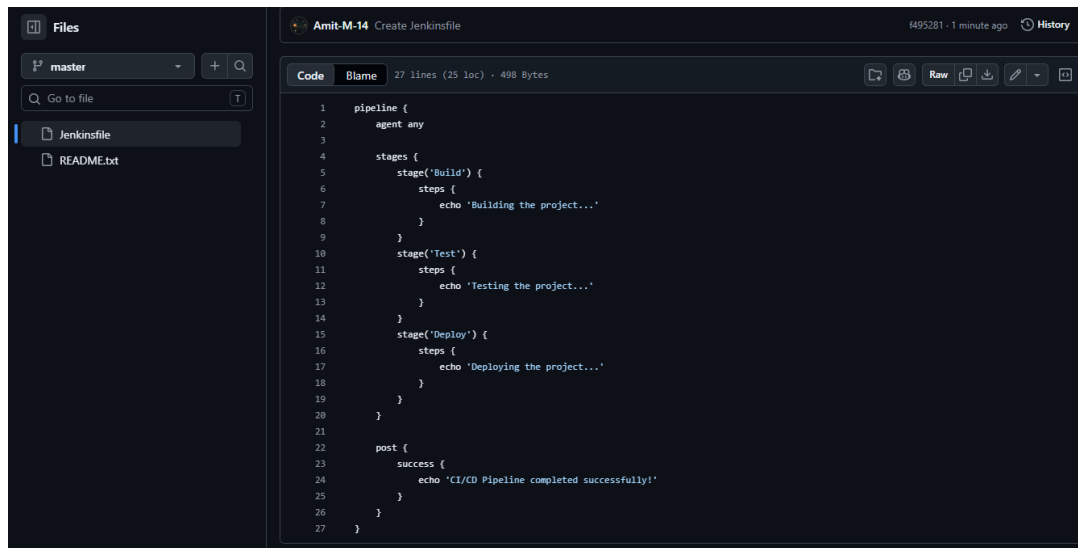
Jenkinsfile

☒ Lightweight checkout ?

Evidence 16: GitHub repository containing the Jenkinsfile.



Evidence 17: Jenkinsfile stored in GitHub, defining Build, Test and Deploy stages.



```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Build') {
6       steps {
7         echo 'Building the project...'
8       }
9     }
10    stage('Test') {
11      steps {
12        echo 'Testing the project...'
13      }
14    }
15    stage('Deploy') {
16      steps {
17        echo 'Deploying the project...'
18      }
19    }
20  }
21
22  post {
23    success {
24      echo 'CI/CD Pipeline completed successfully!'
25    }
26  }
27 }
```

Evidence 18: Jenkins Build #2 status showing the Git revision and repository.

Jenkins / devops-git-lab-pipeline / #2

Status

Changes

Console Output

Delete build '#2'

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

✓ #2 (Sep 10, 2026, 1:42:17 AM)

Started by user [Amit Mohanty](#)

git

Revision: f49528176c3f8d1b9e42362915a7b6d5d607dc5
Repository: <https://github.com/Amit-M-14/devops-git-lab>

- refs/remotes/origin/master

</>

No changes.

Add description

Keep this build forever

Started 13 min ago
Took 12 sec

localhost:8080

REST API Jenkins 2.568.3

Evidence 19: Jenkins Console Output — GitHub checkout and Jenkinsfile retrieval.

```
Started by user Amit Mohanty
Obtained Jenkinsfile from git https://github.com/Amit-M-14/devops-git-lab.
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\devops-git-lab-pipeline\.git #
timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/Amit-M-14/devops-git-lab. # timeout=10
Fetching upstream changes from https://github.com/Amit-M-14/devops-git-lab.
> git.exe --version # timeout=10
> git --version # 'git version 2.53.0.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/Amit-M-14/devops-git-lab.
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision f49528176c3f8d1b9e42362915a7b6d5d607cfc5 (refs/remotes/origin/master)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f f49528176c3f8d1b9e42362915a7b6d5d607cfc5 # timeout=10
Commit message: "Create Jenkinsfile"
```

Evidence 20: Jenkins Console Output — Build, Test, Deploy stages and Finished: SUCCESS.

```
Testing the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Evidence 21: Build Time Trend showing successful builds #1 and #2.



5. Observations

1. Git was verified and configured for the project (username and email).
2. README.txt was tracked and committed on the master branch.
3. A feature-update branch was created, modified and committed separately.
4. The feature branch was merged into master successfully using a fast-forward merge.
5. The local repository was connected to the GitHub remote and both master and feature-update were pushed.
6. Jenkins successfully retrieved the repository from GitHub using Git SCM.
7. The Jenkinsfile defines Build, Test and Deploy stages in a declarative pipeline.
8. Jenkins executed the pipeline successfully and the console output ended with a successful build status.

Conclusion

The required Git, GitHub and Jenkins evidence has been recorded. The successful Jenkins build demonstrates that source code can be retrieved from GitHub and processed through an automated Build-Test-Deploy pipeline.